Agentic AI

1. Courses

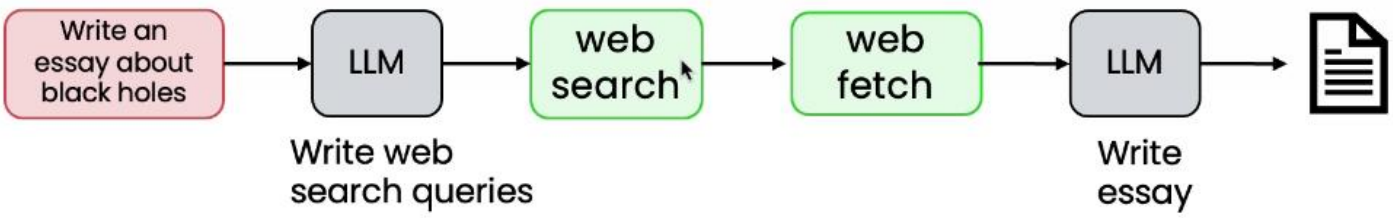
1.1. Agentic AI - Deep Learning.AI Course

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1.1.1 Module 1: Introduction to Agentic Workflows

Error analysis and evals.
What is Agentic AI: Non agentic workflow (zero-shot) is prompt an LLM and receive the response. For example, the user prompts the LLM with a request: write an essay, and the LLM writes it from start to finish.
Agentic workflow is iterative , akint too thinking and doing research (is the human in the loop?). Write, revise, in loop. It takes longer, but it delivers a better result.
An agentic AI workflow is a process where an LLM based app executes multiple steps to complete a task. The LLM decides whether to continue or to stop.
The big task must be broken down into smaller steps so that an agent can execute them. The course builds a resource agent. Uses Tavily to search the web How autonomous Agents are?
Degrees of autonomy: there are different degrees to which systems can be agentic.
Some agents can be less autonomous:

- Maybe some operations are hardcoded, so the agent writes web search queries, but the steps are already hardcodd.
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- Red: user input
- Gray: calls to the LLM
- Green boxes: steps where an external software is used to execute the step

Less autonomous when: steps are predermined, tools are hard coded, autonomy is in text generation (in the example of the essay). So the agent is onyl autonomous when it calls the LLM.

An agent can be more autonomous: they can deide the tool to use, how to use the results to call another tool , etc.
Highly autonomous when: the agent makes mamny decisions autonomously, the agent can create new tools on the fly.
In between there are semi-autonomous agents.
Benefits: many tasks can be executed at once, modularity. Agents make better use of the underlying LLM, using reflection, and other techniquest. Using the agent improves the results, much more that the improvements due to the LLM itself, for example from GPT-3.5 to GPT-4. An agent, for example can perform web search in multiple instances in parallel. Each web search in turns, can have for example 3 LLM(s) that do the web fetch. So, for example, 3 parallel LLM(s) doing the web search, each of them has 3 LLM(s) doing the web fetch. Finally, all results are summarized to write an essay.
So, look at each individual component, like the LLM, and or swap components. So, the system must be modular to allow to swap components later. So, key benefits of agentic workflows are:

- Much better performance,
- Faster than humans because of parallelization
- Modularity, it must be possible toupdate tools, swap out models

1.1.1.1 Agentic AI applications

- Invoice processing: extract the most important fields from the invoice and record them in a database The LLM must also detect whether it is an actual invoice or not. The agent can transform PDF to structured text, and update the DB. This is a pipeline, there is a clear flow.
- Responding to a customer email. A customer received a wrong product, but he want the right one soon. Extract information, find relevant customer record and draft a response for human review.
- A more ambitious agent may be able to reply to more geneal queries from a customer. Example: do you have black blue jeans or blue jeans? The agent must plan the sequence of queries to the database. In these kind of applications, the required steps are not known ahead of time.
- Computer used by agents: the agent has control over the desktop and can open applications, etc. Current agents still struggle in understanding browsers and performing tasks like booking a flight.-

What are the tasks an agent is must suited to?

- If there is a clear, step-by-step process, there are standard procedures to follow. Tasks are easier, with text only assets.
- Harder tasks are when steps are not known ahead, procedures are not standard, and data is encoded in unstructured formats. For example in case it needs multimodal data, like sound, etc-

The most important process is figuring out what are the individual steps to be automatized

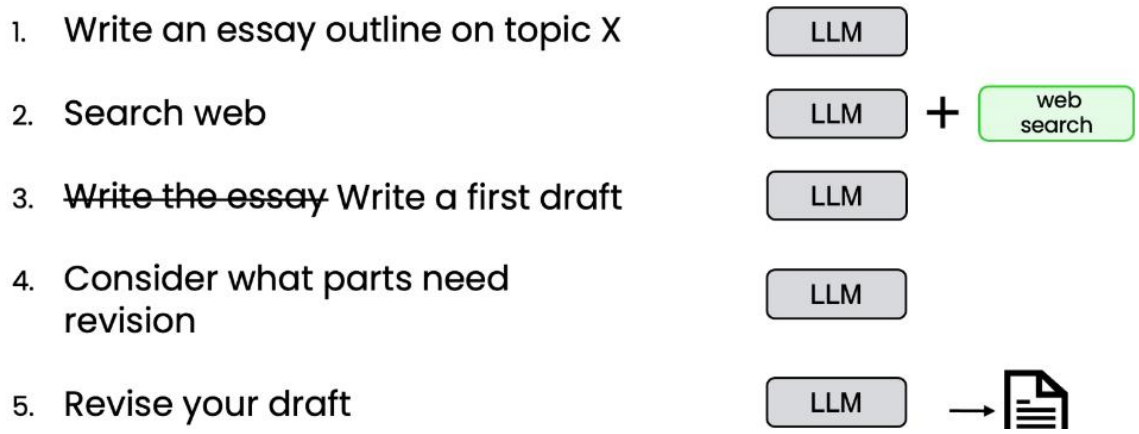
1.1.1.2 Task decomposition - Identifying the steps in a workflow

People do a lot of stuff. Break the stuff into discrete steps.
If the task is: write an essay on topic X, the steps are:

- Promt an LLM and generate the text

The workflow can be
- Write an outline
- Search the web
- Use the web search results to write the essay

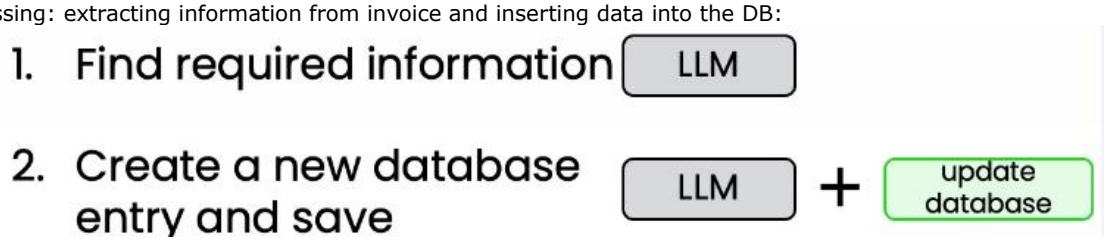
Look at each step and ask whether the step can be done by code of by an LLM. In this case: write the outline can be done by the LLM, the search the web step can be done by an LLM that generates the web seatch queries. Write the essay is also a task for an LLM. When trying to write the essay, an LLM may determine that the essay results are not good enough.
What would you as human do, if essay is not good? Cosider what parts need to be revised and revise the draft.



Other example: responding to the customer email:



Invoice processing: extracting information from invoice and inserting data into the DB:



When building an agent workflow, think: *What building blocks do I have?*

Building block	Examples	Use cases
Models	LLMs	Text generation, tool use, information extraction
	Other AI models	PDF-to-text, text-to-speech, image analysis
Tools	API	Web search, get real-time data, send email, check calendar,....
	Information retrieval	Databases, Retrieval Augmented Generation (RAG)
	Code execution	Basic calculator, data analysis

You must be able to evaluate your agentic workflow.

1.1.1.3 Evaluating agentic AI (evals)

One of the biggers predictor of success is driving a disciplined evaluation process. Evals. After building an agentic WF. What are the things that can go wrong. Run the agent, inspect the results of each step and check for things that you wish it was done better. For example, you inspect the email sent to the customer and it mentions a competitor, which is not good. To do that, manually inspect the result. In that case, to track progress on this issue, build an eval, to track how ofter this error occurs.
For example create a list of competitors: ComCp, Rival Co, The Other Co, etc. It can be done with a tet search no need to use an LLM. Checking the produced text using a text search is an objective method, it is possible to write a program to do that. But LLM(s) can produce non structured text, so an LLM can be used as a judge. A technique can be that the judge LLM assigns a score to the result (example of the essay, but there are better techniques to do that).
Evaluating Agentic AI

- Use code (objective evals) or LLM-as-a-judge (subjective evals).
- Two types f evals: end-to-end and component level.
- Examine traces to perform error analysis
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